

The formula with irrational roots

ANSWER KEY • FOR TEACHERS



Interactive version

REMEMBER

Square roots and the formula

Level 1 • threshold

g8.sqrt

1.

Which is $\sqrt{48}$ in its simplest form?

(A) $16\sqrt{3}$

(B) $2\sqrt{12}$

(C) $6\sqrt{8}$

(D) $4\sqrt{3} - 48 = 16 \cdot 3,$
so $\sqrt{48} = \sqrt{16 \cdot 3}.$
 $\sqrt{3} = 4\sqrt{3}.$

Level 1 • threshold

g9.qe.6

2.

Solve $2x^2 - 5x + 2 = 0$ with the formula.

Answer: $x_1 = \frac{1}{2}, x_2 = 2$

Level 1 • threshold

g9.qe.2

3.

Solve $(x + 1)^2 = 6$. Write the roots exactly: type $\sqrt{}$ as sqrt and $\pm$ as +-, or give the two roots separated by a semicolon.

Answer: $x = -1 \pm \sqrt{6}$

WORKED EXAMPLES

Worked examples

YOUR TURN

Level 2 • satisfactory

g9.qe.16

Solve $x^2 + 2x - 7 = 0$. Give the roots exactly, in their simplest form.

WORKED SOLUTION

$$D = 32 \text{ and } \sqrt{32} = 4\sqrt{2}, \text{ so } x_{1,2} = \frac{-2 \pm 4\sqrt{2}}{2} = -1 \pm 2\sqrt{2}.$$

Answer: $x = -1 \pm 2\sqrt{2}$

YOUR TURN

Level 2 · satisfactory

g9.qe.16

Solve $3x^2 - 4x - 2 = 0$. Give the roots exactly, in their simplest form.

WORKED SOLUTION

$D = 40$ and $\sqrt{40} = 2\sqrt{10}$, so $x_{1,2} = \frac{4 \pm 2\sqrt{10}}{6}$. The numbers 4, 2 and 6 have the common factor 2:
 $x_{1,2} = \frac{2 \pm \sqrt{10}}{3}$.

Answer: $x = \frac{2 \pm \sqrt{10}}{3}$

CHECK

Check yourself

Level 1 · threshold

g9.qe.16

1.

Simplify $\frac{6 \pm \sqrt{12}}{2}$.

A $3 \pm \sqrt{12}$

B $3 \pm \sqrt{3} - \sqrt{12} = 2\sqrt{3}$, so $\frac{6 \pm 2\sqrt{3}}{2} = 3 \pm \sqrt{3}$: both terms are divided by 2.

C $3 \pm \sqrt{6}$

D $3 \pm 2\sqrt{3}$

Level 2 · satisfactory

g9.qe.16

2.

Solve $x^2 - 2x - 2 = 0$. Give the roots exactly.

WORKED SOLUTION

$D = 12$ and $\sqrt{12} = 2\sqrt{3}$, so $x_{1,2} = \frac{2 \pm 2\sqrt{3}}{2} = 1 \pm \sqrt{3}$.

Answer: $x = 1 \pm \sqrt{3}$

3.

Solve $4x^2 - 4x - 1 = 0$. Show your working, and write the roots exactly, in their simplest form.

MARKING SCHEME: TICK WHAT YOUR SOLUTION HAS

$$D = (-4)^2 - 4 \cdot 4 \cdot (-1) = 32. \quad 1 \text{ pt}$$

$$\sqrt{32} = 4\sqrt{2}, \text{ so } x_{1,2} = \frac{4 \pm 4\sqrt{2}}{8}. \quad 1 \text{ pt}$$

$$\text{Divides all three numbers by 4: } x_{1,2} = \frac{1 \pm \sqrt{2}}{2}. \text{ (Not } 1 \pm \sqrt{2}, \text{ and not } \frac{1 \pm 4\sqrt{2}}{2}.) \quad 1 \text{ pt}$$

WORKED SOLUTION

$$D = (-4)^2 - 4 \cdot 4 \cdot (-1) = 16 + 16 = 32, \text{ and } \sqrt{32} = \sqrt{16 \cdot 2} = 4\sqrt{2}.$$

$$x_{1,2} = \frac{4 \pm 4\sqrt{2}}{2 \cdot 4} = \frac{4 \pm 4\sqrt{2}}{8}. \text{ The numbers 4, 4 and 8 have the common factor 4:}$$

$$\text{Answer: } x_{1,2} = \frac{1 \pm \sqrt{2}}{2}, \text{ that is, } x_1 = \frac{1 - \sqrt{2}}{2} \text{ and } x_2 = \frac{1 + \sqrt{2}}{2}.$$

4.

Solve $x^2 + 8x + 10 = 0$. Give the roots exactly.

WORKED SOLUTION

$$D = 24 \text{ and } \sqrt{24} = 2\sqrt{6}, \text{ so } x_{1,2} = \frac{-8 \pm 2\sqrt{6}}{2} = -4 \pm \sqrt{6}.$$

$$\text{Answer: } x = -4 \pm \sqrt{6}$$

5.

Above, you found that $x = 1 + \sqrt{3}$ is a root of $x^2 - 2x - 2 = 0$. Show it by substituting: work out the left side exactly, without a calculator.

MARKING SCHEME: TICK WHAT YOUR SOLUTION HAS

$$(1 + \sqrt{3})^2 = 1 + 2\sqrt{3} + 3 = 4 + 2\sqrt{3}. \quad 1 \text{ pt}$$

$$4 + 2\sqrt{3} - 2(1 + \sqrt{3}) - 2 = 4 + 2\sqrt{3} - 2 - 2\sqrt{3} - 2 = 0, \text{ so it is a root.} \quad 1 \text{ pt}$$

WORKED SOLUTION

$$(1 + \sqrt{3})^2 = 1 + 2\sqrt{3} + 3 = 4 + 2\sqrt{3}.$$

$$\text{So } (1 + \sqrt{3})^2 - 2(1 + \sqrt{3}) - 2 = 4 + 2\sqrt{3} - 2 - 2\sqrt{3} - 2 = 0.$$

The left side is exactly 0, so $1 + \sqrt{3}$ is a root. With a rounded value such as 2.73 the left side would only be close to 0.

EXAM TASK

Exact roots without a calculator

Level 3 • basic

PUPP style • 3 points

g9.qe.16

The equation $x^2 - 10x + 18 = 0$ is given.

- Show that its discriminant is 28.
- Show that its roots are $5 - \sqrt{7}$ and $5 + \sqrt{7}$.
- Between which two consecutive whole numbers does each root lie? Explain without a calculator.

MARKING SCHEME: TICK WHAT YOUR SOLUTION HAS

- $D = (-10)^2 - 4 \cdot 1 \cdot 18 = 100 - 72 = 28$. 1 pt
- $\sqrt{28} = 2\sqrt{7}$, so $x_{1,2} = \frac{10 \pm 2\sqrt{7}}{2} = 5 \pm \sqrt{7}$. 1 pt
- $2 < \sqrt{7} < 3$ (because $4 < 7 < 9$), so $2 < 5 - \sqrt{7} < 3$ and $7 < 5 + \sqrt{7} < 8$. 1 pt

WORKED SOLUTION

- $D = (-10)^2 - 4 \cdot 1 \cdot 18 = 100 - 72 = 28$.
- $\sqrt{28} = \sqrt{4 \cdot 7} = 2\sqrt{7}$, so $x_{1,2} = \frac{10 \pm 2\sqrt{7}}{2} = 5 \pm \sqrt{7}$.
- $4 < 7 < 9$, so $2 < \sqrt{7} < 3$. Then
 - $5 - 3 < 5 - \sqrt{7} < 5 - 2$, that is, $2 < 5 - \sqrt{7} < 3$;
 - $5 + 2 < 5 + \sqrt{7} < 5 + 3$, that is, $7 < 5 + \sqrt{7} < 8$.

Answer: c) the root $5 - \sqrt{7}$ lies between 2 and 3, and the root $5 + \sqrt{7}$ between 7 and 8.

TEACHER NOTES

For the teacher

Pupils can say: “I can solve a quadratic equation whose roots are irrational, write the roots exactly and in their simplest form, and round them when that is asked for.”

Where this lesson sits. Lesson 1.3 used the formula only when D was a perfect square. Here the same formula meets any positive D . The new work is grade 8’s simplifying of roots (31.1.1) used inside the formula, and cancelling a common factor of the **whole** numerator and the denominator. The second garden from lesson 1.3 is solved in the first frame. The cases $D = 0$ and $D < 0$ are lesson 1.5.

Common mistakes and what helps

- Dividing only one term of the numerator: $\frac{4 \pm 2\sqrt{5}}{2} = 2 \pm 2\sqrt{5}$ (t1, c1–c4). The numeric check $\frac{6 \pm 4}{2} = 5$ convinces most pupils; so does taking the factor out first, $\frac{2(2 \pm \sqrt{5})}{2}$.
- “Halving under the root”: $\frac{\sqrt{12}}{2} = \sqrt{6}$ (c1).
- Cancelling the wrong factor when $a \neq 1$: $\frac{6 \pm 2\sqrt{3}}{4}$ becomes $\frac{3 \pm \sqrt{3}}{2}$, not $3 \pm \sqrt{3}$ (Example 2, t2, c3). Ask “which number divides all three?”
- Rounding too early: rounding $\sqrt{129}$ to 11 gives a width of 4 m and an area of 28 m², not 30 m².
- Taking $\sqrt{D}$ out incompletely ($\sqrt{32} = 2\sqrt{8}$) (r1).

About the answer boxes. A box accepts any exact form with the right value, also an unsimplified one such as $(4 + \sqrt{32})/8$. After a right answer, the feedback shows the simplest form. The core item c3 is therefore a written task, marked on the simplest form. In class, ask for the simplest form every time. Pupils type $\pm$ as +-; the last frame of Explain says so.

Key terms (EN • LT): irrational number • *iracionalusis skaičius*; exact value • *tiksloji reikšmė*; approximate value • *apytikslė reikšmė*; to take a factor out of the root • *iškelti daugiklį prieš šaknies ženklą*; common factor • *bendrasis daugiklis*.

Timing (45 min):

- Remember: 5 min (the Refresh box on simplifying roots may be needed).
- Hook: 3 min.
- Explain: 10 min.
- Examples 1–2 with their “Your turn”: 12 min.
- Check c1–c3 (the core; c3 is written): 10 min.
- Summary: 2 min.

c4 (more practice), c5 (checking an irrational root by substituting it, level 3) and the exam task are for pupils who finish early, or for homework.

Programme: 32.1.1.1 “*Įrodoma ir taikoma kvadratinės lygties sprendinių formulė.*” It rests on 31.1.1 (grade 8) “*Praktikuojamasi įkelti teigiamą skaičių į pošaknį ir iškelti jį prieš šaknies ženklą.*” The PUPP items “find D , then the roots” of 2025 and 2026 all had a perfect-square D ; this lesson prepares for the general case. The Unit 1 check and the autumn test each have an equation with irrational roots. Achievement A1.